

Renan Alves de Oliveira

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Professional Summary

I enjoy solving complex problems, whether analyzing data from the cosmos or simply coding a solution to a problem. My strong foundation in physics and computational modeling has enabled me to transform raw data into value. Currently, I design and develop a multi-agent platform with an easy, user-friendly interface. My expertise ranges from developing robust ETL/ELT data pipelines to deploying fine-tuned LLMs for specific domains.

Work Experience

AI Specialist

May 2025 – Present

Turing

Campinas, SP

- Architecting production multi-agent system on AWS managing 200+ specialized agents and 1K+ API integrations (LangChain, LlamaIndex, n8n, WSO2) with RAG, web search, and third-party services to automate complex backoffice and service optimization workflows at scale.

AI Specialist

January 2025 – Present

Psyche Aerospace

Campinas, SP

- Developed computer vision pipeline processing multispectral drone and satellite imagery to compute vegetation indices (NDVI among others) for automated crop health assessment and precision recommendations.
- Built AWS infrastructure (EC2, SageMaker, Bedrock, EKS) to automate web scraping and fine-tuning of Microsoft Phi-4 on large-scale agricultural corpus, creating domain-specific LLM for agribusiness.
- Implemented high-performance RAG system using GPU-accelerated FAISS (FLAT index) over 3M+ documents, achieving 95-100% recall with low latency for hallucination detection and real-time information retrieval in production.

Project Leader

September 2023 – December 2024

Samsung Electronics

Campinas, SP

- Led end-to-end development of advanced fitness metrics for Samsung Galaxy Watch (VO2max/%VO2max, AT/AnT, HRmax estimations), coordinating cross-functional team to deliver features to millions of active users.
- Coordinated sensor data integration (HR, accelerometer, GPS) and algorithm validation with research teams, maintaining technical documentation and achieving on-time delivery rate across product releases.

Data Engineer Sr

August 2022 – December 2024

Samsung Electronics

Campinas, SP

- Built automated ETL pipelines (using Pandas, PostgreSQL) for processing time-series health metrics (HR, speed, VO2/VCO2), improving model training efficiency and reducing manual preprocessing time.
- Implemented data quality framework with automated validation and error-handling in CI/CD (GitHub Actions), reducing production data errors and ensuring consistent data integrity across environments.
- Redesigned data ingestion and storage architecture (MinIO, PostgreSQL) with engineering team, improving data retrieval speed and enabling real-time dashboards in PowerBI.

Researcher

August 2018 – April 2024

Universidade Federal do Espírito Santo

Vitória, ES

- Developed and deployed an interactive dashboard (Streamlit, later React/TypeScript) with a structured database for astronomical data visualization, enabling researchers to efficiently analyze complex datasets from a remote server. Accelerated student research through technical mentorship in Python, Unix, and Shell, while optimizing computational workflows with Cython, Numba, and parallelization techniques. Managed HPC cluster operations to ensure optimal uptime and resource allocation for critical scientific computations.

Researcher Analyst

November 2019 – May 2020

Simons Foundation (Flatiron Institute)

New York, NY

- Developed a high-performance neural emulator (PyTorch, deployed via ONNX to OpenVINO) to accelerate dark matter simulations, enabling single-GPU predictions of particle evolution using a StyleGAN-based architecture for parameter conditioning. Validated the model's statistical accuracy (correlation and transfer functions) against semi-analytical and standard methods. Presented accepted paper at NeurIPS 2020 Machine Learning and the Physical Sciences workshop.

Researcher

August 2016 – July 2018

Universidade Estadual de Londrina

Londrina, PR

- Validated non-Gaussianity and isotropy assumptions in cosmological models through statistical analysis (Pearson's χ^2 test) of Planck satellite Cosmic Microwave Background data. To facilitate this work, developed polyMV, a Python/C software library for efficient conversion of spherical harmonic coefficients to Cartesian vector representations, with applications beyond cosmology.

Junior Researcher

May 2014 – July 2014

Hydrologic Research Center

Del Mar, CA

- Modeled critical failure conditions for landslide initiation through geotechnical slope stability analysis, applying continuum mechanics and numerical methods as part of a Science Without Borders internship.

Education

Doctor in Astrophysics, Cosmology and Gravitation

August 2018 – April 2024

Universidade Federal do Espírito Santo

Vitória, ES

- Thesis: *Probing cosmology with an eye on Rubin: from strong lensing to the large scale structure of the universe*

Master in Physics

August 2016 – July 2018

Universidade Estadual de Londrina

Londrina, PR

- Dissertation: *Testes de isotropia utilizando vetores de multipolo em pequenas escalas*

Bachelor of Science in Applied Physics

July 2013 – May 2014

California State University, San Marcos

San Marcos, CA

- *Science Without Borders Program*

Bachelor in Physics

March 2010 – March 2016

Universidade Estadual de Londrina

Londrina, PR

Technical Skills

AI & Machine Learning: Agents, LLM Fine-tuning, Multimodal Prompting, ONNX, PyTorch, RAG

Programming: C, Cython, Numba, Python, R, TypeScript

Cloud & Infrastructure: AWS, CI/CD Pipelines, Docker, GitHub Actions

Data Engineering: Artifactory, Data Pipelines, DVC, ETL, MinIO, MLflow, pgAdmin, Pinecone, PostgreSQL, Zilliz

Version Control & Collaboration: Confluence, Code Review, Git, GitHub, GitLab, Jira

Data Analysis: Bayesian Inference, Meta-Analysis, Monte Carlo Simulation, Statistical Testing

Visualization & Reporting: Power BI, Streamlit, React, Scientific Writing, DataViz

HPC & Optimization: Code Optimization, High-Performance Computing, Parallel Computing

Selected Publications

- **Alves de Oliveira, R.** et al. "The Last Stand Before Rubin: consolidated sample of strong lensing systems". *arXiv preprint*, 2025. arXiv:2509.09798
- **Alves de Oliveira, R.** et al. "Field-level neural network emulator for cosmological n-body simulations". *The Astrophysical Journal*, 2023. 10.3847/1538-4357/acdb6c
- **Alves de Oliveira, R.** et al. "CMB statistical isotropy confirmation at all scales using multipole vectors". *Physics of the Dark Universe*, 2020. doi:10.1016/j.dark.2020.100608

Languages & Certifications

Languages: Portuguese (Native), English (Full Professional Proficiency)

Certifications: Data Science (Unicamp)